6. If A is square matrix of order 3 then $|kA|$ is

(a) $k|A|$ (b) $-k|A|$

(c) $k^3|A|$ (d) $-k^3|A|$

7. $adj(AB)$ is equal to

(a) $adjA\ adjB$ (b) $adjA^T\ adjB^T$

(c) $adjB\ adjA$ (d) $adjB^T\ adjA^T$

8. The inverse matrix of $\begin{pmatrix} \frac{4}{5} & \frac{-5}{12} \\ \frac{-2}{5} & \frac{1}{2} \end{pmatrix}$ is

(a) $\frac{7}{30}\begin{pmatrix} \frac{1}{2} & \frac{5}{12} \\ \frac{2}{5} & \frac{4}{5} \end{pmatrix}$ (b) $\frac{7}{30}\begin{pmatrix} \frac{1}{2} & \frac{-5}{12} \\ \frac{-2}{5} & \frac{1}{5} \end{pmatrix}$

(c) $\frac{30}{7}\begin{pmatrix} \frac{1}{2} & \frac{5}{12} \\ \frac{2}{5} & \frac{4}{5} \end{pmatrix}$ (d) $\frac{30}{7}\begin{pmatrix} \frac{1}{2} & \frac{-5}{12} \\ \frac{-2}{5} & \frac{4}{5} \end{pmatrix}$

9. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that $ad - bc \neq 0$ then A^{-1} is

(a) $\frac{1}{ad-bc}\begin{pmatrix} d & b \\ -c & a \end{pmatrix}$

(b) $\frac{1}{ad-bc}\begin{pmatrix} d & b \\ c & a \end{pmatrix}$

(c) $\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

(d) $\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ c & a \end{pmatrix}$

10. The number of Hawkins-Simon conditions for the viability of an input-output analysis is

(a) 1 (b) 3

(c) 4 (d) 2

11. The inventor of input - output analysis is

(a) Sir Francis Galton

(b) Fisher

(c) Prof. Wassily W. Leontief

(d) Arthur Caylay

12. Which of the following matrix has no inverse

(a) $\begin{pmatrix} -1 & 1 \\ 1 & -4 \end{pmatrix}$ (b) $\begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix}$

(c) $\begin{pmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{pmatrix}$

(d) $\begin{pmatrix} \sin a & \sin a \\ -\cos a & \cos a \end{pmatrix}$

13. The inverse matrix of $\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ is

(a) $\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$ (b) $\begin{pmatrix} -2 & 5 \\ 1 & -3 \end{pmatrix}$

(c) $\begin{pmatrix} 3 & -1 \\ -5 & -3 \end{pmatrix}$ (d) $\begin{pmatrix} -3 & 5 \\ 1 & -2 \end{pmatrix}$

14. If $A = \begin{pmatrix} -1 & 2 \\ 1 & -4 \end{pmatrix}$ then $A(adjA)$ is

(a) $\begin{pmatrix} -4 & -2 \\ -1 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} 4 & -2 \\ -1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ (d) $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$

15. If A and B non-singular matrix then, which of the following is incorrect?

(a) $A^2 = I$ implies $A^{-1} = A$

(b) $I^{-1} = I$

(c) If $AX = B$ then $X = B^{-1}A$

(d) If A is square matrix of order 3 then $|adj\ A| = |A|^2$

16. The value of $\begin{vmatrix} 5 & 5 & 5 \\ 4x & 4y & 4z \\ -3x & -3y & -3z \end{vmatrix}$ is

(a) 5 (b) 4 (c) 0 (d) –3

17. If A is an invertible matrix of order 2 then $det(A^{-1})$ be equal to

(a) $det(A)$ (b) $\frac{1}{det(A)}$

(c) 1 (d) 0

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